

Assignment-1

MVC Design Patterns and its Variants

Overview

The model-View-Controller (MVC) design pattern divides an application into three interconnected components:

- Model: Manages the data and business logic.
 - View: Handles the presentation layer and user interface.
 - Controller: processes user input and interacts with the model and view.
- The separation facilitates modular development, easier testing and maintainance.

Variants of MVC

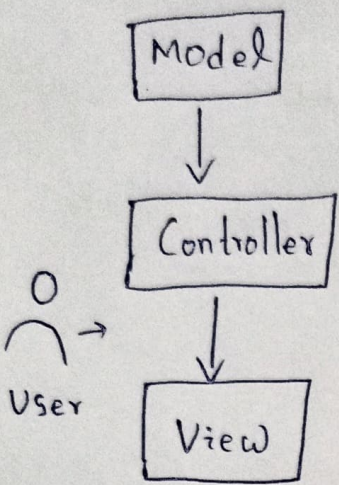
1. Model-view-Presenter (MVP)

- Presenter: Acts as an intermediary between the view and model, handling all presentation logic.
- Use case: suitable for applications requiring clear separation between UI and business logic, such as Windows Forms application.

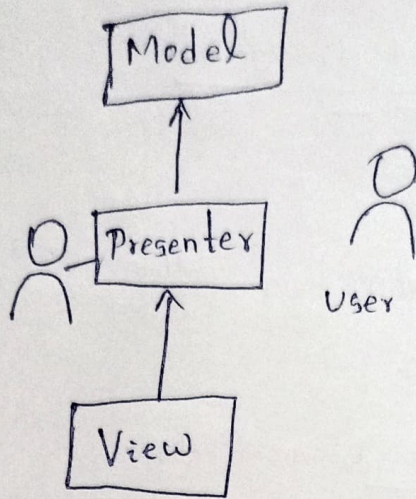
2. Model-view-ViewModel (MVVM)

- view model: Exposes data and commands from the model to the view, facilitating data binding.
- Use case: ideal for applications using data-binding capabilities, like WPF or Angular application.

MVC



MVP



MVVM

